Null and alternative hypotheses

1 A man is arrested for the crime of stealing from his employer and will be tried in court.

a Write the null hypothesis for this situation in words.

b Write the alternative hypothesis for this situation in words.

2 It is known that projector bulbs have a mean lamp life of 4000 hours. Lampland claim that their latest bulb has a mean lamp life of more than 4000 hours.
What set of hypotheses, H_0 and H_1, should Lampland consider in order to demonstrate their claim is correct?

3 What set of hypotheses, H_0 and H_1, should a quality control officer inspecting a flour factory consider, in order to demonstrate that $500\ \mathrm{g}$ bags of flour are meeting quality standards?

4 The public transport department of a small city wishes to determine whether introducing bus lanes will improve travelling times. The mean travel time for a bus between two locations is known to be 33 minutes. A bus lane is constructed and new travel times are recorded.
What set of hypotheses, H_0 and H_1, should the department consider as part of its decision process to determine whether the bus lane is justified?

5 A cruise company suspects that the average percentage of female customers is increasing and wishes to redirect its marketing strategy accordingly. They calculate the average percentage of female passengers for the past financial year and compare it to data collected five years ago where the average percentage of female passengers was found to be 59%.
What set of hypotheses, H_0 and H_1, should the cruise company use in order to determine if their suspicion is correct?

6 A researcher is interested in estimating the average number of children per family in a certain large city. The researcher randomly samples 200 family units and determines a sample mean $\overline{x} = 2.37$. She is concerned that $\overline{x}$ is different to the government gazetted mean of 2.5 children per family for the entire city.

a Write the null hypothesis, H_0, for this situation.

b Write the alternative hypothesis, H_1 for this situation.

7 A teacher is interested in the average weekly income of students in Year 12. The teacher randomly samples 20 students and determines a sample mean $\overline{x} = \$45.40$ per week. He is fairly confident that the government reported mean of $\$37.50$ per week is too low and should be revised.

a Write the null hypothesis, H_0, for this situation.

b Write the alternative hypothesis, H_1 for this situation.

8 A manufacturer has a machine that produces light bulbs continuously on a production line. He tests a batch of 12 randomly chosen bulbs to see how long they last when lit. He finds that the average life is 2035 hours. The manufacturer is thinking about increasing the claimed average of 2000 hours referred to on each of the cardboard light bulb boxes, but needs a sound statistical argument to support the change.

a Write the null hypothesis, H_0, for this situation.

b Write the alternative hypothesis, H_1, for this situation.

9 A water dispenser used for scientific experiments should deliver 200 ml of water to a plastic cup each time a dispensing button is pressed. As it is a scientific instrument, the manufacturer plans to check the device by drawing a sample batch of 30 cups of water and calculating a sample mean. The manufacturer suspects that the machine is delivering less than 200 ml each time.
What set of hypotheses should the manufacturer consider?

10 A state football coach is interested in finding out if the mean Yo-Yo test scores of country athletes, μ_1, is higher than city athletes, μ_2. He decides to conduct a two sample t-test in order to investigate.
Write the null hypothesis and alternative hypothesis for this test.

11 Mrs Jones is interested in how many hours a week her students spend on social media and whether there is a difference in the mean amount of time spent by boys, μ_1, compared to the mean amount of time spent by girls, μ_2. She surveys her students and conducts a two sample t-test.

a State the null hypothesis in words.

b State the null hypothesis using symbolic notation.

c State the alternative hypothesis in words.

d State the alternative hypothesis using symbolic notation.

12 The colours in a packet of smarties are said to be evenly distributed by the manufacturer. Emma opens a packet of 60 smarties and finds the distribution as follows:

Pink	Blue	Red	Yellow	Orange	Brown	Purple	Green
6	8	10	7	6	6	9	8

Emma decides to test if this sample is consistent with the manufacturer's claim using a χ^2 goodness of fit test.

a Write down the null hypothesis for the test in words.

b Write down the alternative hypothesis for the test in words.

13 A student wished to investigate how often teachers exercised. She decides to conduct a χ^2 test of independence to see whether amount of exercise is independent of subject area. She recorded the teacher's subject area and the number of minutes of exercise per week in the table below:

	$0-60$	$60-180$	$180-360$	$360+$
Maths	1	3	3	1
Science	4	2	2	0
English	2	3	1	0
Physical education	0	1	3	4

State the null hypothesis for this study in words.